

Research Methods — Module 06: Learning & Psychiatry — Study Questions

Part 1: Fundamental Comprehension

1. **The Computational Shift:** Define computational psychiatry. How does it fundamentally differ from the descriptive, symptom-cluster approach of the Diagnostic and Statistical Manual of Mental Disorders (DSM)?
2. **Dysfunction Taxonomy:** Active Inference reframes mental disorders as specific computational dysfunctions. Define and contrast the four main archetypes: precision dysregulation, structural model deficits, aberrant prior preferences, and updating failures.
3. **The Aberrant Precision Account:** Detail the Active Inference explanation for hallucinations and delusions in schizophrenia. How does an imbalance in precision-weighting between top-down priors and bottom-up prediction errors generate psychosis?
4. **Modeling Depression:** How does Active Inference conceptualize major depressive disorder? Specifically, explain how "pessimistic priors" and reduced precision on positive prediction errors mathematically mandate the lived experience of anhedonia and learned helplessness.
5. **The Threat Model:** Explain the computational mechanics of clinical anxiety. How does elevated precision assigned to threat-related prediction errors trap the generative model in a state of chronic hypervigilance?
6. **The Autistic Spectrum:** Describe Lawson's Active Inference account of autism. How does a consistently high precision applied to low-level sensory prediction errors explain both the autistic individual's intense sensory overwhelm and their extraordinary attention to detail?

Part 2: Advanced Analysis & Synthesis

1. **The Pharmacological Bridge:** What is "pharmacological validation" in computational modeling? Detail how specific neuromodulators (e.g., dopamine, serotonin,

noradrenaline) map to distinct mathematical parameters (like precision or learning rate) within an Active Inference generative model.

2. **Comorbidity Mechanics:** A patient is diagnosed with both severe anxiety and depression. Instead of viewing these as separate diseases, how does the Active Inference framework explain their co-occurrence as a single, complex failure mode of a unified generative model?
3. **Treatment Resistance:** Why do some patients fail to respond to traditional SSRIs or cognitive behavioral therapy? Explain treatment resistance through the lens of a highly precise, structurally rigid generative model that actively explains away therapeutic interventions.
4. **Critique of the Paradigm:** Critically evaluate the current state of computational psychiatry. Is the field genuinely ready to improve individual patient outcomes in the clinic, or is it currently serving only as a sophisticated post-hoc descriptive language? Justify your stance.

Part 3: Methodological Application

1. **Digital Phenotyping Design:** Design a smartphone-based behavioral task involving a probabilistic reversal learning game. What specific Active Inference parameters (e.g., volatility estimation, sensory precision) would this game extract from the user's tap behaviors over two weeks?
2. **Differential Diagnosis Modeling:** You have two patients diagnosed with generic "depression" under the DSM. Design a computational experiment (combining a behavioral task and a generative model tracking their responses) that could mathematically distinguish if Patient A is suffering from an updating failure while Patient B is suffering from aberrant prior preferences.
3. **Adaptive Clinical Trials:** Explain how a Bayesian adaptive clinical trial utilizing Active Inference principles differs from a classical randomized controlled trial (RCT). How does continuous computational phenotyping enable "trial enrichment"?
4. **Translational Pipeline Execution:** Outline the five-step translational pipeline required to move a purely theoretical Active Inference model of a psychiatric disorder into an FDA-approved clinical diagnostic tool.

5. **Ethical Implications:** What are the primary ethical risks of deploying continuous computational phenotyping (e.g., scraping smartphone usage patterns to infer hidden states regarding depressive relapse) in a vulnerable clinical population? Draft a consent policy to mitigate these risks.